

Programme	B. Sc. Computer Science				
Course Title	Computer Hardware Assembly				
Type of Course	Minor				
Semester	I				
Academic Level	100 - 199				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	Basic understanding of computer operation No previous experience in hardware assembly required				
Course Summary	Students will learn about the different components of a computer system, how they work together, and the skills necessary to assemble and maintain computer hardware effectively				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	A comprehensive understanding of fundamental concepts in Computer Organization and Hardware	U	C	Instructor-created exams / Assignment
CO2	Students will be able to understand and identify computer hardware components	Ap	P	Viva Voce
CO3	Students will be able to proficiently assemble computer hardware components adhering to industry standards and best practices.	C	P	Practical / Group Work
CO4	Students will learn to install and configure various operating systems (e.g., Windows, Linux) and drivers on	C	C	Practical / Group Work

	newly assembled computer systems			
CO5	Students will acquire the skills to diagnose and troubleshoot common hardware issues encountered during computer assembly	E	P	Practical/Exam/ Assignments
CO6	Students will develop the skills to perform hardware upgrades	C	P	Practical / Group Work
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs	Marks
I	Basic Computer Organization and Concept of Hardware		11	17
	1	Basic Computer Organization: Input Unit, Storage Unit, Processing Unit, Control Unit, Output Unit	1	
	2	CPU Architecture: Arithmetic and Logic Unit, Control unit, Registers	1	
	3	Memory: Primary Memory, Secondary Memory	1	
	4	Access Time, Storage Capacity-bit, byte, nibble	1	
	5	Cache memory, Primary Memory- RAM (Static , Dynamic), ROM	2	
	6	Secondary Memory, storage devices (Magnetic tape, Hard disk , SSD and CD drive). Memory hierarchy.	3	
	7	Input and Output Devices	2	
II	Hardware Components		12	18
	8	Concept of Hardware and Software	1	
	9	Microprocessor, Clock Speed and Performance, Types of processors (Single core, dual core, multi core), GPU	2	
	10	Inside CPU: SMPS, Motherboard, Processor, Storage Devices (HDD, SSD), RAM (DDR2, DDR3, DDR4), ROM	2	
	11	Motherboard Components: Processor Slot, Cooling Fan, RAM, Expansion Slots (PCIe), Mouse and Keyboard Ports, Chipset, BIOS/UEFI Chip, SATA/NVMe Slots, Network Interface, Ports- Ethernet, VGA port, HDMI port, USB port	3	

	12	Cables and Connectors,	2	
	12	Expansion Cards: Graphics card, Sound Card, Network Interface Card	1	
III	Hardware Assembling		10	17
	14	Safety and Tools: Introduction to ESD (Electrostatic Discharge) safety, use of antistatic wrist straps, and essential tools for assembling hardware	1	
	15	Assembling a PC: Step-by-step guide on assembling a PC, including installing processor onto the motherboard, Setup Cooling Fan, Install RAM, Install other expansion cards, Mounting the motherboard into the case and Install storage devices (HDD, SSD) into drive bays, Install the GPU into the appropriate PCIe slot (if not integrated into the CPU)	4	
	16	Cable Management: Best practices for managing cables within a PC case to ensure optimal airflow and aesthetics. Connect power supply cables to the motherboard, CPU, GPU, and storage devices, Connect case cables (power switch, reset switch, LEDs, USB ports) to the appropriate headers on the motherboard,	3	
IV	System Configuration, OS Installation, trouble Shooting		12	18
	17	BIOS and UEFI: Understanding the roles of BIOS and UEFI navigating BIOS settings, and configuring hardware.	2	
	18	Installing and Configuring Operating Systems: Guidelines for installing operating systems (Windows, Linux)	2	
	19	Installing Drivers: install drivers for motherboard components (Chipset, LAN, Audio), GPU, and other peripherals.	2	
	20	Hardware Upgrades: How to upgrade components such as RAM, storage, and GPUs, including compatibility considerations.	2	
	21	Troubleshooting Common Assembly Issues: Identifying and resolving common issues encountered during PC assembly	2	
	22	Diagnostics and Maintenance: Introduction to methods, tools and software used for diagnosing hardware issues.	2	
V	Hands-on Hardware Assembling Practical Applications, Case Study and Course Project		30	

	1	1: Identifying Computer Components ● Identify and describe the function of the CPU, RAM, motherboard, PSU, storage devices, and peripheral connectors. ● Disassemble and reassemble a desktop computer, identifying each component as it is removed and replaced. 2: Building a PC from Scratch ● Use appropriate tools and safety equipment to assemble a computer, including installing the motherboard, CPU, CPU cooler, RAM, and storage. ● Practice cable management to ensure a neat and efficient build. ● Document each step of the assembly process for future reference and learning. 3: Operating System Installation and Configuration ● Install a chosen operating system (e.g., Windows, Linux) from a bootable USB drive or DVD. ● Install essential drivers and software updates. ● Configure basic settings (user accounts, network settings, display resolution). Case Study: Trouble Shooting and Maintenance	25	
			5	

References

1. Pradeep K. Sinha and Priti Sinha, Computer Fundamentals: Concepts, Systems & Applications. BPB Publications.
2. Bigelow's Troubleshooting, Maintaining & Repairing PCs Hardcover – by Stephen Bigelow
3. Kevin Wilson, Computer Hardware: The Illustrated Guide to Understanding Computer Hardware. Amazon Digital Services LLC – KDP, 2018.

Mapping of COs with PSOs and POs:

	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6
CO 1	1	2	-	-	-	-						
CO 2	-	2	-	-	-	-						
CO 3	-	2	-	-	1	-						
CO 4	-	2	-	-	-	-						
CO 5	-	2	-	-	1	-						
CO 6	-	2	-	-	1	-						

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Project Evaluation	End Semester Examinations
CO 1	✓	✓		✓

CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6	✓	✓		✓